THALLURU LAKSHMI SAI KALYAN

DATA ENGINEER

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PROFESSIONAL SUMMARY

Data Engineer with 6.5+ years of total professional IT experience, including nearly 4+ years of hands-on experience in the Hadoop Ecosystem, designing and developing scalable, high-performance data applications using PySpark, Hive, and SQL. Skilled in building and optimizing real-time and batch data pipelines, data ingestion frameworks, and cloud-based big data solutions (AWS, Azure) within Agile delivery environments. Proven ability to partner with cross-functional teams to translate business requirements into reliable, production-grade data engineering solutions.

KEY ACHIEVEMENTS

- Designed and managed real-time payment data pipelines (Bill Payments, ACH, Card, and Wire Transfers) using Kafka and PySpark, ensuring scalable ingestion and processing across diverse transaction types.
- Developed and deployed fraud detection models leveraging historical claims and payment patterns to identify anomalies such as duplicate claims, velocity attacks, and geographic inconsistencies.
- Optimized PySpark jobs through partition tuning, broadcast joins, and efficient use of repartition/coalesce, reducing processing latency and improving throughput for 10GB+ datasets.
- Automated pipeline orchestration and monitoring with Kafka checkpoints and Spark Structured Streaming, and integrated Prometheus/Grafana dashboards for reliability and transparency.
- Collaborated on release cycles by integrating updated models and schema changes into production workflows with minimal downtime.

TECHNICAL SKILLS

Big Data & Distributed Processing: Apache Spark, Apache Hive, Apache Kafka, HDFS, Hadoop Ecosystem

Languages & Scripting: Python, Scala, SQL, Shell Scripting

Cloud Platforms: AWS (S3), Microsoft Azure

Orchestration, Scheduling & Monitoring: Apache Airflow, AutoSys

Data Formats: Parquet, ORC, AVRO, JSON, XML, Sequence Files, Text

Domain Expertise: Experienced in the banking sector, specializing in real-time payment processing (Bill Payments, ACH, Card, Wire Transfers) and developing fraud detection models

Methodologies & Tools: Agile, Scrum, Waterfall, SDLC, Jira

PROFESSIONAL EXPERIENCE

I.T. Analyst — **Oct 2024 – Present**

Tata Consultancy Services — Bank of America

- Executed end-to-end Agile project delivery, including impact analysis, data mapping, solution design, and code deployment for enterprise data engineering initiatives.
- Designed, developed, and maintained scalable data pipelines and data ingestion processes for large-scale financial datasets.
- Implemented data transformations and aggregations aligned to business logic using Spark, Hive, and Shell Scripting.
- Reduced job runtime through performance tuning of PySpark scripts and Spark jobs.

- Provided production support, root-cause analysis, and defect resolution on a priority basis to ensure system reliability.
- Evaluated and standardized Data Engineering tools and frameworks to improve productivity and output quality across the team.
- Led end-to-end application migration projects, ensuring data integrity and minimal service disruption.
- Monitored and troubleshot data pipelines to ensure seamless, uninterrupted data flow.
- Partnered with clients to gather requirements and communicate technical solutions effectively.
- Designed and implemented a Deposit Model for fraud labelling and protection.
- Filtered records by handling RDI Claims, Loss Data Match, Account Closure data, and Fraud classification.
- Generated fraud protected deposit features to strengthen downstream analytical and ML models.
- Automated data filtering and classification to improve fraud detection accuracy and efficiency.
- Hands-on experience in the banking domain, building real-time payment processing pipelines (Bill Payments, ACH, Card, Wire Transfers) and deploying fraud detection models to enhance transaction security

Programming Analyst — **Nov 2020 – Jul 2023**

Cognizant — H.E.B Projects

- Ingested data from HDFS into Apache Spark and built DataFrames to analyze data across multiple staging layers.
- Developed Scala-based Spark code to seamlessly read data from HDFS and construct DataFrames for downstream processing.
- Flattened complex JSON data using Spark APIs to create structured DataFrames.
- Reconstructed complex JSON and XML structures to meet downstream business requirements.
- Built multi-stage Spark data transformation pipelines to support business and analytics needs.
- Authored high-level estimates and design documents for new software functionality.
- Worked extensively with Text, Parquet, ORC, Sequence, AVRO, and JSON file formats.
- Provided ongoing production support, including priority-based issue resolution and defect management.
- Performed data cleansing and transformation to support accurate downstream analytics.
- Collaborated with QA teams to troubleshoot testing issues and supplied test data as needed.
- Analyzed data and stored output datasets on Amazon S3 for downstream consumption.
- Implemented new features and enhancements based on customer requirements.
- Conducted code and peer reviews, unit testing, and bug fixing for assigned tasks.
- Prepared and delivered daily and weekly status reports to customers on project activities.
- Participated in daily Scrum meetings to track progress on projects and assigned tasks.

Programming Analyst Trainee — **Dec 2018 – Nov 2020**

Cognizant — MetLife Projects

- Created HDFS directories and Hive tables to support structured data storage.
- Worked with dental insurance coverage datasets, performing data cleansing and transformation.
- Loaded processed data into Hive tables for downstream reporting and analysis.
- Loaded and transformed large volumes of semi-structured data, including XML, JSON, Avro, and Parquet.
- Read and wrote data to/from AWS S3 using Apache Spark.
- Performed code and peer reviews, unit testing, and bug fixing for assigned tasks.

EDUCATION

Bachelor of Engineering- *R.M.D Engineering College* — **Aug 2014 – Jun 2018**